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ForceVLA2: Unleashing Hybrid Force-Position Control with Force Awareness for Contact-Rich Manipulation

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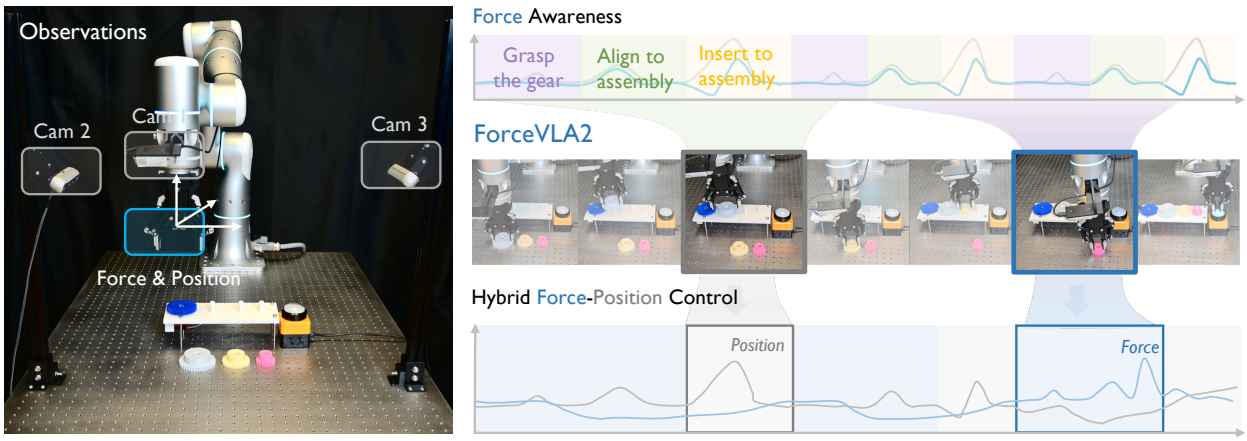


Figure 1. **ForceVLA2 concept.** Contact-rich manipulation requires force regulation, beyond visual and state observations (left). **ForceVLA2** integrates force information across multiple scales, enabling rich modeling of contact dynamics. It builds force awareness into task planning through incoming force signals, and it outputs hybrid force–position actions with dynamic balance (right).

Abstract

Embodied intelligence for contact-rich manipulation has predominantly relied on position control, while explicit awareness and regulation of interaction forces remain under-explored, limiting stability, precision, and robustness in real-world tasks. We propose ForceVLA2, an end-to-end vision–language–action framework that equips robots with hybrid force-position control and explicit force awareness. ForceVLA2 introduces force-based prompts into the VLM expert to construct force-aware task concepts across stages, and employs a Cross-Scale Mixture-of-Experts (MoE) in the action expert to adaptively fuse these concepts with real-time interaction forces for closed-loop hybrid force–position regulation. To support learning and evaluation, we construct ForceVLA2-Dataset, containing 1,000 trajectories over 5 contact-rich tasks, including wiping, pressing, and assembling, with multi-view images, task prompts, proprioceptive state, and force signals. Extensive experiments show that ForceVLA2 substantially improves success rates and reliability in contact-rich manipulation, outperforming π_0 and $\pi_{0.5}$ by 48.0% and 35.0%, respectively, across the 5 tasks, and mitigating common failure modes such as arm overload and unstable contact,thereby actively advancing force-aware interactive physical intelligence in VLAs. The project page is available at this URL.

1. Introduction

Artificial intelligence has advanced rapidly with large vision–language models (VLMs) [1, 6, 35, 42, 44, 46] that exhibit remarkable perceptual and reasoning capabilities. However, these models remain confined to virtual domains, lacking the embodiment necessary for authentic physical understanding and interaction in real-world settings. To overcome this limitation, vision–language–action models (VLAs) [2, 3, 22, 47, 50] extend VLMs toward physical intelligence by seamlessly connecting perception and reasoning to embodied interaction. This integration unifies cognition and manipulation into a cohesive framework, enabling robots not only to perceive and understand, but also to manipulate objects in the physical world in ways that closely mirror human intelligence. Consequently, VLAs have be-

come a foundation for robotic learning, paving the way for enhanced sample efficiency, generalizability, compositionality, and incremental learning [21], and enabling adaptive, context-aware behavior in dynamic, complex environments.

While recent VLAs such as OpenVLA [22], OpenVLA-OFT [23], GR00T-N1 [2], and π_0 [3] have advanced embodied cognition by linking perception and language to action, these models demonstrate strong performance in semantic grounding and language following, showing that pretrained multimodal models can acquire priors and be efficiently fine-tuned with only a few demonstrations. However, current VLAs still lack the ability to reason about physical dynamics and fine-grained contact interactions, which are essential for real-world, contact-rich manipulation. Building upon π_0 [3], ForceVLA [43] extends this framework by incorporating the force modality, leading to improved success rates in contact-rich manipulation tasks.

Yet, existing VLAs reduce force to an auxiliary perceptual input rather than leveraging it for active, adaptive closed-loop force interaction, a core requirement for genuine physical intelligence. In contrast, humans rely on high-level visual and linguistic reasoning to identify stage-specific targets and estimate coarse spatial relations, while integrating real-time force sensing as direct physical perception to adaptively refine both force and position during interaction. Current VLAs, however, still lack mechanisms to reason about the spatiotemporal relationships required across different task stages and to integrate force perception with active force–position interaction within subtasks. This gap motivates a unified treatment of *force awareness* and *closed-loop force–position control* within the VLA paradigm for contact-rich manipulation.

To overcome these limitations, we propose **`ForceVLA2`**, a novel framework that equips VLAs with active hybrid force-position control with force awareness to enhance contact-rich manipulation, as shown in Fig. 1. By incorporating force modality into the VLA architecture, ForceVLA2 links language-level and visual semantic reasoning with physical contact-level interaction, enabling physically grounded, generalizable, and highly adaptive behaviors, especially in tasks requiring fast and stable force-following. Inspired by the human motor control system, where force perception and regulation are hierarchically integrated [16], ForceVLA2 employs a dual-level design. A long-horizon reasoning module driven by a force prompt is embedded into the VLM expert to construct force-aware task concepts, while a short-horizon reactive manipulation skill employs a Cross-Scale Mixture-of-Experts (MoE) to integrate force-aware task knowledge from the VLM expert with embodied interaction forces in the action expert, enabling adaptive regulation of hybrid force–position interaction. Extensive experiments demonstrate that ForceVLA2 consistently improves performance and generalization in contact-rich manipulation. Our contributions are summarized as follows:

- We introduce ForceVLA2, the first end-to-end hybrid force–position control framework with force awareness for VLAs, enhancing contact-rich manipulation by integrating force-prompt-driven VLM reasoning and fusing force into action experts via a Cross-scale MoE, enabling coordinated hybrid force–position interaction and tightly linking VLM reasoning with action generation to improve physical interaction.
- We construct ForceVLA2-Dataset, a dataset of 1,000 trajectories across 5 contact-rich tasks, including wiping, pressing, and assembling, with fundamental modalities such as multi-view images, task prompts, and proprioceptive states. It is further enhanced by adding force prompts and incorporating force into the action space, bridging the gap between embodied hybrid force–position interaction and force-aware task concepts, providing an empirical foundation for advancing research from force perception to interaction.
- We conducted extensive experiments demonstrating the effectiveness of ForceVLA2, compared to π_0 and $\pi_{0.5}$, the success rates increased by 48.0% and 35.0%, respectively, across the 5 tasks, represents notable progress toward bridging force-aware task concept with embodied adaptive hybrid force–position manipulation skills.

2. Related Work

Vision-language-action models. Recent research on VLAs [4, 5, 10, 20, 22, 23, 26, 33, 48] leverages large-scale multimodal pretraining to generalize robotic policies across diverse tasks and embodiments, mapping visual observations and language instructions to low-level actions through end-to-end learning. Autoregressive models [22, 23, 29] treat control as next-token prediction over discretized action tokens, enabling VLM-style training with efficient decoding. Diffusion-based approaches [7, 12, 27, 34, 51] model continuous action distributions for more diverse and controllable behaviors, though at high computational cost. Flow-based methods such as π_0 [3] accelerate action generation via flow-matching decoders, while ForceVLA [43] extends it with force inputs to improve contact-rich manipulation accuracy. Dual-system designs like $\pi_{0.5}$ and GR00T-N1 [2] separate high-level reasoning from low-level action generation, combining generalization with action.

Physical modality integration in robot policies. Traditional VLA models struggle with dynamic, contact-rich interactions that demand fine-grained feedback, motivating recent efforts to integrate force and tactile modalities for improved control. Recent works incorporating force sensing [15, 17, 28, 40, 43] enhance motion stability and contact precision, yet force is typically used only as a perceptual cue rather than an active control signal. Tactile-based methods such as TLA [13], Tac-Man [49], and other multimodal fu-

sion approaches [18, 24, 52] improve robustness under occlusion but cannot substitute for direct force sensing, since tactile-to-force estimation remains indirect and noisy. Despite these advances, existing methods remain limited to force perception and purely position-based interaction.

Multimodal embodied datasets. Large-scale embodied datasets have significantly advanced robot learning, yet they rarely include standardized force modality, which is essential for modeling contact [31, 36, 38]. Open-X-Embodiment [8] aggregates over one million trajectories across 22 robots, though only a subset of its datasets include force modalities, leading to sparse and inconsistent coverage. Recent datasets have begun to incorporate explicit force modality for contact-rich manipulation. RH20T [11] provides multi-modal data, including RGB-D, audio, proprioceptive state, and calibrated force collected via haptic teleoperation, while REASSEMBLE [32] focuses on assembly tasks with synchronized visual, proprioceptive, and force modalities, incorporating sub-tasks to enable detailed analysis of manipulation skills.

3. ForceVLA2 Framework

ForceVLA2 is designed to unify force awareness and hybrid force–position control within a vision–language–action (VLA) framework, thereby enabling precise, adaptive, and stable contact-rich manipulation. Inspired by findings in human sensorimotor control, we posit that force acts as a unifying signal across the perception–planning–execution hierarchy, providing essential feedback for dynamic and compliant interaction [16, 37]. Neuroscientific evidence further supports this perspective, showing that humans perceive external forces through proprioceptive sensing and regulate them via internal forward models implemented in the cerebellum and motor cortex [16]. Motivated by biological sensing–acting mechanisms, ForceVLA2 internalizes multi-scale force cues within both context-dependent long-horizon reasoning, realized through a force prompt, and adaptive, stage-aware reactive skills, enabling precise force–position interaction for unified, force-aware manipulation that bridges perception and action.

Building upon these principles, we propose the ForceVLA2 architecture, which integrates multi-scale perception, contextual reasoning, and force-aware manipulation into a unified VLA framework. As shown in Fig. 2, multi-view images, task prompts, and force prompts are jointly encoded within a VLM, providing high-level guidance for task operation sequencing and spatial–semantic reasoning in task decomposition (Sec. 3.1). In parallel, force observation bypasses high-level fusion and modulates the action expert via a direct gradient pathway, enabling a reactive response to observed force during contact (Sec. 3.2.1). Force and proprioceptive tokens are routed through a Cross-Scale Mixture-of-Experts (MoE) [30], which dynamically integrates visual, language, and force information to determine the dominant modality at each phase of the task (Sec. 3.2.2). ForceVLA2 unifies force–position control and task progress prediction into a cohesive, force-aware closed-loop architecture, enabling physically grounded reasoning and adaptive perception–action coordination for contact-rich manipulation.

3.1. Long-Horizon Force Awareness via Prompting

Current VLA models rely on pre-trained priors for task decomposition; however, purely vision–language inputs are insufficient in contact-rich settings that require human-like task awareness and physically grounded reasoning about task concepts, particularly interaction forces [9]. In such contact-rich scenarios, visual cues are often ambiguous or insufficient, whereas essential information is conveyed through force feedback, as in aerospace connector insertion or gear assembly. To address this, we introduce force prompts as textual cues that indicate the current subtask and encode stage-specific physical context, thereby constructing force-aware task concepts. This mechanism enables ForceVLA2 to inherit VLM knowledge, assess subtask completion, transition across stages, and explicitly update force cues to guide force-aware manipulation.

Vision, task, and force prompt integration. Inspired by the π_0 framework [3], the ForceVLA2 architecture integrates vision, task prompts, and force prompts before a conditional flow-matching action expert. Visual observations from multiple RGB cameras, along with task instructions, are encoded by a SigLIP-based [45] vision–language model (based on PaliGemma [1]) into contextual embeddings. The text prompt provides a global description of the task, while the force prompt encodes the current subtask state. Each task has a predefined list of subtasks, and the force prompt determines whether to maintain the current subtask or transition to the next, functioning like a discrete-state machine.

Specifically, the visual tokens $\mathbf{E}_v \in \mathbb{R}^{N_v \times D_{\text{model}}}$ are first processed through a visual encoder $f(\cdot)$: $\mathbf{Z}_v = f(\mathbf{E}_v) \in \mathbb{R}^{N_v \times D_{\text{model}}}$. The linguistic tokens, obtained by concatenating the text prompt $\mathbf{T}_t \in \mathbb{R}^{N_t \times D_{\text{model}}}$ and the force prompt $\mathbf{T}_f \in \mathbb{R}^{N_f \times D_{\text{model}}}$, are processed through a text encoder $g(\cdot)$: $\mathbf{Z}_l = g([\mathbf{T}_t; \mathbf{T}_f]) \in \mathbb{R}^{(N_t+N_f) \times D_{\text{model}}}$, where N_t and N_f are the number of text and force prompt tokens. The outputs of the visual and text encoders are concatenated to form the input to the vision–language model: $\mathbf{E}_{\text{in}} = [\mathbf{Z}_v; \mathbf{Z}_l] \in \mathbb{R}^{(N_v+N_t+N_f) \times D_{\text{model}}}$. Finally, $\mathbf{E}_{\text{in}}$ undergoes token-to-token attentions within the vision–language model ($\texttt{VLM}(\cdot)$), producing the fused contextual tokens:

$$\mathbf{E} = \texttt{VLM}(\mathbf{E}_{\text{in}}) \in \mathbb{R}^{(N_v+N_t+N_f) \times D_{\text{model}}}. \quad (1)$$

The resulting sequence $\mathbf{E}$ serves as the fused multi-modal representation, which is subsequently used for downstream short-term force perception and control.

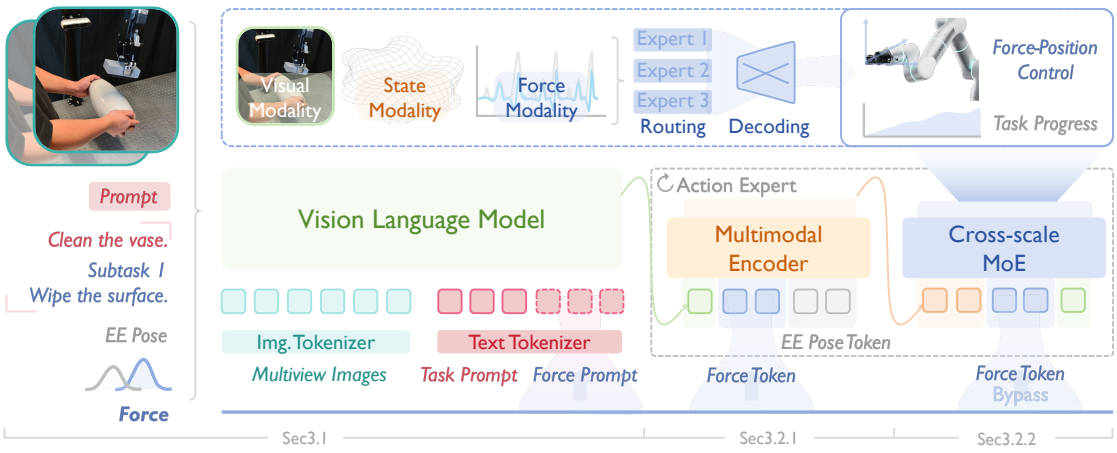


Figure 2. **Framework of ForceVLA2.** ForceVLA2 takes multi-view images, task and force prompts, and proprioceptive states (EE pose and force) as input. Force is injected at multiple scales: as sub-task prompts fused with images in the vision-language model, and as force tokens combined with EE pose in the multimodal encoder, with a bypass to preserve raw signals. The cross-scale MoE integrates these modalities to produce hybrid force–position actions and track sub-task progress for adaptive, contact-rich manipulation.

3.2. Short-Horizon Force-to-Control Loop

Existing VLA models rely on end-effector (EE) 6D pose control, which suffices for coarse tasks but fails when subtle force interactions dominate. ForceVLA2 encodes proprioceptive state and force differently within the action expert, passing them through the multi-modal encoder to fuse with visual–language embeddings for long-horizon reasoning, while transient force signals bypass the fusion layer and feed directly to the MoE, forming a short-term reactive loop. This design preserves gradient fidelity for rapid force feedback, avoids over-reliance on past trajectories, and enables active exploration (Appendix A). The MoE dynamically integrates long- and short-term cues to produce joint predictions of EE 6D pose, force, and subtask progress for hybrid force-position control in contact-rich manipulation.

3.2.1. multi-modal Encoding for EE 6D pose and Force

Proprioceptive-force encoding. The proprioceptive state, represented by the EE 6D pose, is expressed as $\mathbf{p} \in \mathbb{R}^7$ and encoded through another linear layer ϕ_P:

$$\mathbf{E}_P = \phi_P(\mathbf{p}) \in \mathbb{R}^{D_{\text{model}}}. \tag{2}$$

To enable force-aware perception, the raw 6D force–torque readings $\mathbf{f}_{\text{raw}} \in \mathbb{R}^6$ are first projected into the model embedding space via a linear mapping ϕ_F:

$$\mathbf{E}_F = \phi_F(\mathbf{f}_{\text{raw}}) \in \mathbb{R}^{D_{\text{model}}}. \tag{3}$$

The encoded EE 6D pose and force tokens are concatenated to form a multi-modal state representation, $\mathbf{E}_{\text{state}} = [\mathbf{E}_P; \mathbf{E}_F]$, which is then fed into the MoE alongside visual–language contextual tokens $\mathbf{E} \in \mathbb{R}^{N_{\text{VL}} \times D_{\text{model}}}$. Meanwhile, a force embedding $\mathbf{E}_F$ bypasses the cross-modal interaction and is sent directly to the MoE, forming a short-term, reactive pathway.

Cross-modal interaction. To incorporate high-level task context into proprioceptive sensing, the state embeddings interact with the vision–language features through cross-attention. Specifically, the encoded visual–language sequence $\mathbf{E} \in \mathbb{R}^{N_{\text{VL}} \times D_{\text{model}}}$ serves as the contextual key–value pair, while the state tokens act as queries:

$$\mathbf{E}'_{\text{state}} = \texttt{CrossAttn}(\mathbf{E}_{\text{state}}, \mathbf{E}). \tag{4}$$

This operation injects global task semantics into the local proprioceptive streams, allowing the model to interpret force and motion signals under the guidance of visual–language intent. The conditioned embeddings are then concatenated as:

$$\mathbf{E}_{\text{cond}} = [\mathbf{E}; \mathbf{E}'_{\text{state}}; \mathbf{E}_F], \tag{5}$$

which forms the multi-modal input for the following expert routing stage.

3.2.2. Adaptive Routing and Decoding

Cross-scale Mixture-of-Experts. The conditioned sequence $\mathbf{E}_{\text{cond}}$ is processed by a sparse MoE layer designed to adaptively select the most informative modality for the current manipulation phase. The MoE contains three modality-specific experts (visual, state, and force), each implemented as a lightweight MLP [14]. A dynamic gating network computes token-wise routing weights $\mathbf{w} = [w_V, w_S, w_F]$, and activates the most relevant expert for each token:

$$\mathbf{E}_{\text{MoE}} = \sum_{m \in \{V,S,F\}} w_m \cdot \texttt{Expert}_m(\mathbf{E}_{\text{cond}}). \tag{6}$$

Through this mechanism, the network autonomously emphasizes visual reasoning during free-space motion and force or position cues when contact and compliance become critical. The fused representation $\mathbf{E}_{\text{MoE}}$ is then projected to

the latent space of the flow-matching policy [25] for downstream action generation.

Flow Matching policy for position-force control. The flow-matching policy head generates force-aware actions by progressively denoising a noise-initialized action conditioned on the fused multi-modal context. At the beginning of each time step t, a noisy action sample $\mathbf{a}_t^{(0)} \sim \mathcal{N}(0, I)$ is iteratively refined according to the learned conditional flow:

$$\frac{d\mathbf{a}_t^{(\tau)}}{d\tau} = F_\theta\big(\mathbf{a}_t^{(\tau)}, \mathbf{E}_{\text{MoE}}, \tau\big), \tau \in [0, 1], \tag{7}$$

where τ is the denoising time, $\mathbf{E}_{\text{MoE}}$ provides the contextual multi-modal embedding from the MoE module. In practice, this continuous flow is implemented as a discrete denoising process:

$$\mathbf{a}_t^{(\tau+1)} = \mathbf{a}_t^{(\tau)} + \Delta_\tau \cdot F_\theta\big(\mathbf{a}_t^{(\tau)}, \mathbf{s}_t, \mathbf{E}_{\text{MoE}}\big), \tag{8}$$

starting from random noise and gradually converging to the final force–position command:

$$\mathbf{a}_t = \mathbf{a}_t^{(1)} = [\Delta\mathbf{p}_t; \mathbf{f}_t], \Delta\mathbf{p}_t \in \mathbb{R}^7,\ \mathbf{f}_t \in \mathbb{R}^6,\ s_t \in [0, 1]. \tag{9}$$

Here, $\Delta\mathbf{p}_t$ represents the desired change in end-effector pose, $\mathbf{f}_t$ denotes the predicted contact force. By conditioning the denoising process on the fused visual–language–force representation, the model achieves closed-loop, context-aware control that adapts fluidly to contact-rich interaction.

Probabilistic modeling of force-control transition. Note that the transition indicator s_t is modeled as a continuous floating-point value output by the model, ranging from 0 to 1. Once s_t exceeds 1 (threshold), the controller transitions to the next subtask and resets the value to 0. The ground truth value for s_t during training, denoted as $\hat{s}_t$, is computed based on the observed quantities. Let A denote the current end-effector position and A_t the target position, E be the current end-effector orientation, E_t the target orientation, and $\overrightarrow{AA_t}$ represent the displacement from A to A_t. Define the observed statistics: $\Theta = \frac{1}{2}\left(\frac{\vec{E}\cdot\vec{E_t}}{\|\vec{E}\|\|\vec{E_t}\|} + 1\right) \in (0, 1)$, $L = \|\overrightarrow{AA_t}\| \in (0, \infty)$, $F = \|\mathbf{f}\| \in [n, m]$. Here, Θ measures orientational alignment (1 = aligned, 0 = opposite), L is the remaining distance to the goal, and F represents the instantaneous contact force (bounded within the normalization limits n and m).

We define the ground truth for the subtask transition probability as the joint event that these quantities satisfy their respective progress conditions. Given observed values (θ, l, f) and assuming $\Theta \sim \texttt{Beta}(\alpha, 1)$, $L \sim \texttt{Exponential}(\lambda)$, $F \sim \texttt{Uniform}(n, m)$, we calculate the subtask transition probability $\hat{s}_t$ as follows:

$$\begin{aligned}\hat{s}_t &= P(\Theta \le \theta,\ L \ge l,\ F \le f) \\ &= \int_0^\theta \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\,\theta'^{\,\alpha-1} d\theta' \cdot \int_l^\infty \lambda e^{-\lambda\ell}\, d\ell \cdot \int_n^f \frac{1}{m-n}\, df \\ &= \frac{\Gamma(\alpha+1)}{\alpha\Gamma(\alpha)}\,\theta^\alpha e^{-\lambda l}\,\frac{f-n}{m-n}.\end{aligned} \tag{10}$$

Here, $\Gamma(\cdot)$ represents the Gamma function, with $\alpha = 2$ and $\lambda = 2$. s_t serves as an indicator of subtask completion. When it exceeds a threshold, the controller transitions to the next subtask and resets s_t to 0.

4. ForceVLA2-Dataset

To train the proposed ForceVLA2 model, we collected a new dataset specifically tailored for contact-rich manipulation tasks, with a particular emphasis on capturing force-control signals for sub-task segmentation and annotation (Fig. 3 (a)). Unlike previous work, which primarily relied on EE 6D pose trajectories [8, 11] and VR-based human teleoperation [41], our setup employs the force-feedback GELLO teleoperation framework [39] to capture high-quality demonstrations with natural motion dynamics, while explicitly recording force and motion trajectories. Each manipulation task is segmented into 3–5 annotated subtasks, guided by visual inspection and analysis of the synchronized force signals. The collected dataset constitutes a multi-modal corpus encompassing visual, proprioceptive, task-prompt, force-prompt, and force modalities.

Hardware and sensor setup. Demonstrations were collected using a 7-DOF Flexiv Rizon 4s robotic arm equipped with a DH Robotics AG-95 adaptive gripper. Visual observations were acquired from three RGB cameras: two static third-person views (Intel RealSense D455, 1280×720 @ 30 fps) and one wrist-mounted egocentric view (Intel RealSense D435, 640×480 @ 30 fps), as shown in Fig. 4. A 6D force/torque sensor attached to the end-effector recorded interaction forces at 300 Hz, while the robot joint states and end-effector (EE) 6D poses were logged synchronously. All visual streams were resized to 480×640 resolution, normalized, and timestamp-synchronized prior to storage.

Collected tasks. We curated 5 contact-rich manipulation tasks: *press bottle*, *clean vase*, *clean board*, *retrieve plate*, and *assemble gears* (Fig. 3 (b)). Operators used the force-feedback GELLO teleoperation interface to control the robot in real time, ensuring accurate and repeatable interaction patterns while preserving natural force application. To enhance diversity, object poses are varied across demonstrations. Each task is recorded until completion, and subtask boundaries are annotated offline based on distinct transitions in the force signal profiles.

Dataset statistics. The resulting dataset comprises 1000 trajectories and approximately 500K synchronized timesteps. Each trajectory contains time-aligned streams of

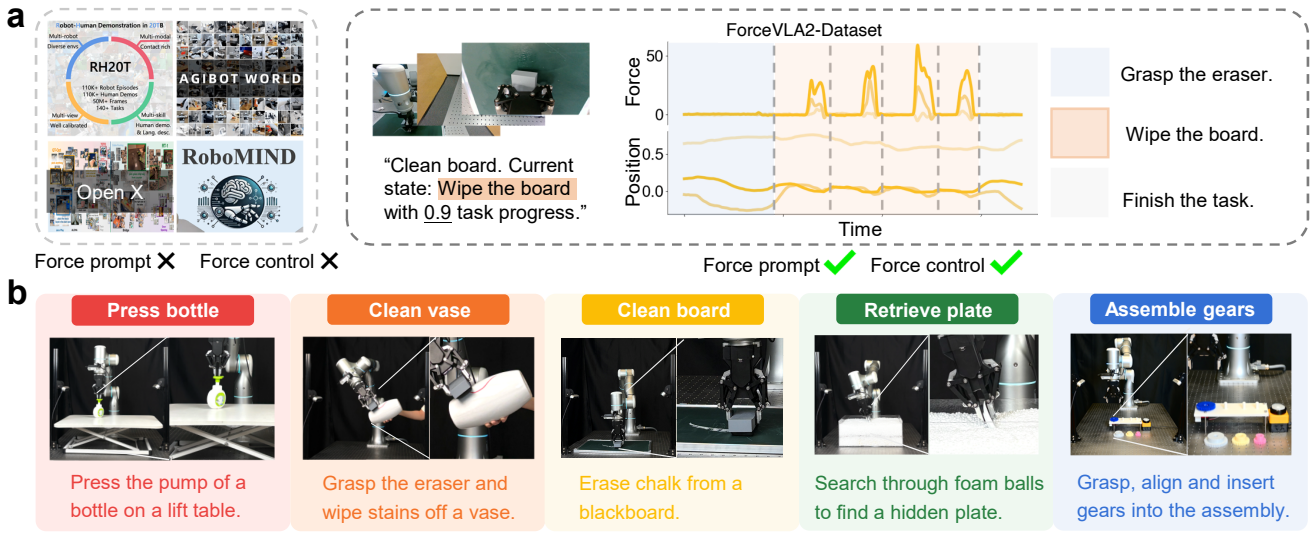


Figure 3. **The illustration of ForceVLA2-Dataset.** (a) ForceVLA2-Dataset is the first dataset with force prompts for task decomposition and the only one providing force-control supervision. (b) It features 1,000 demonstrations across five contact-rich tasks.

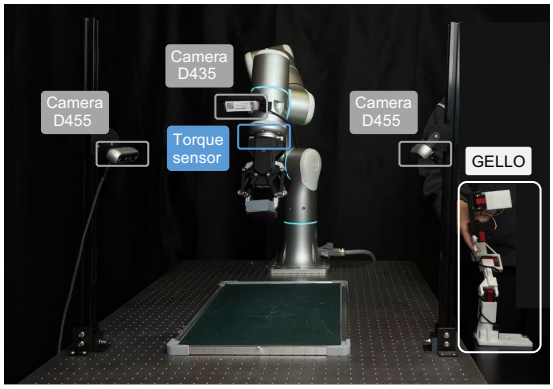


Figure 4. **The dataset collection system.** A Flexiv arm is driven by manually controlled GELLO [39] to accomplish dexterous tasks and record images, force, as well as the pose of the robot.

RGB images, end-effector (EE) 6D poses, and force signals. The ForceVLA2-Dataset facilitates research on force-aware and subtask-structured manipulation policy learning by introducing force prompts and integrating force into the action space, bridging the gap for embodied hybrid force–position interaction and providing an empirical foundation for advancing research from force perception to interaction. The detailed dataset statistics and descriptive information can be found in the Appendix B.

5. Experiments

In this section, we evaluate the effectiveness of ForceVLA2 on diverse contact-rich manipulation tasks that test perceptual and reasoning abilities, as well as adaptive, closed-loop hybrid force–position interactions across multiple task stages under complex physical conditions. The experiments address the following research questions:

- **Q1:** How does ForceVLA2 perform in real-world contact-rich manipulation tasks, and what specific advantages and technical benefits does hybrid force–position control offer over pure position control?
- **Q2:** Which components contribute most significantly to the performance improvements of ForceVLA2?
- **Q3:** Which modality fusion strategy yields the best performance in active hybrid force–position controlling?

5.1. Experiment Setting

Tasks and evaluation metrics. Our experimental benchmark consists of 5 contact-rich manipulation tasks within the proposed ForceVLA2-Dataset: Press the bottle, Clean the vase, Clean the board, Retrieve the plate, and Assemble gears. The primary evaluation metric is the success rate (%), determined by conducting 20 independent trials for each task.

Baselines. To demonstrate the superiority of our proposed ForceVLA2, we compare it against five established methods: ACP [17], π_0 [3], $\pi_{0.5}$ [19], ForceVLA [43] and π_0 with naive force input. ACP leverages principles of admittance control to implicitly incorporate the desired force into the final output by predicting a virtual action. The π_0 model leverages a flow matching architecture integrated with a pre-trained VLM to leverage Internet-scale knowledge. Building upon π_0, the $\pi_{0.5}$ model incorporates co-training on diverse tasks to enhance generalization. Meanwhile, ForceVLA prioritizes external force sensing as a core modality within its VLA architecture.

5.2. Main Experiment Results

Overall performance on contact-rich tasks (Q1). As summarized in Tab. 1, ForceVLA2 significantly outperforms all baselines, achieving a 66% average success rate across all tasks. On force-sensitive tasks, it surpasses the second-best model by up to 50%. Compared with models without force inputs, ForceVLA2 and ForceVLA, which incorporate force feedback, show remarkable improvements

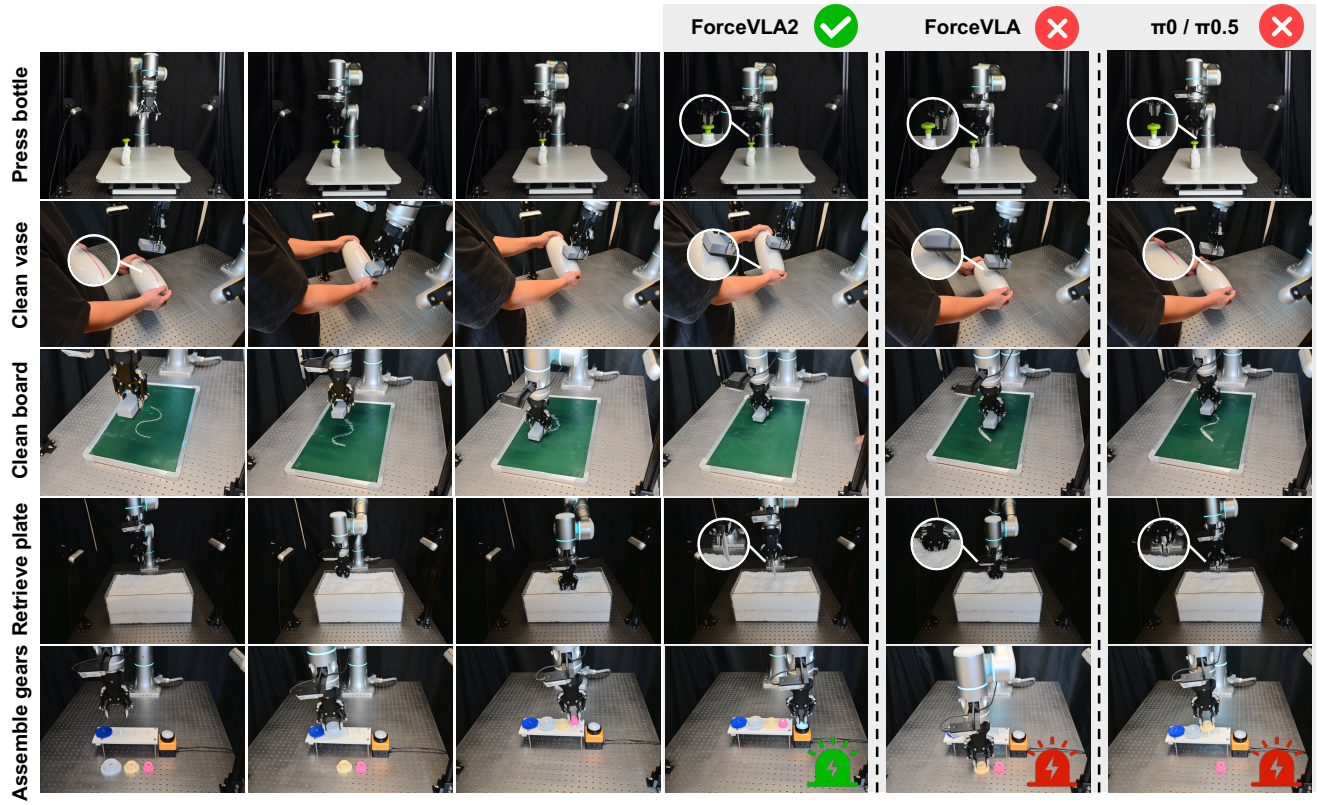


Figure 5. **Qualitative results on typical manipulation tasks (compared with ForceVLA and π serials).** ForceVLA2 completes these tasks with higher success rates and faster execution while avoiding arm overload, demonstrating superior compliance.

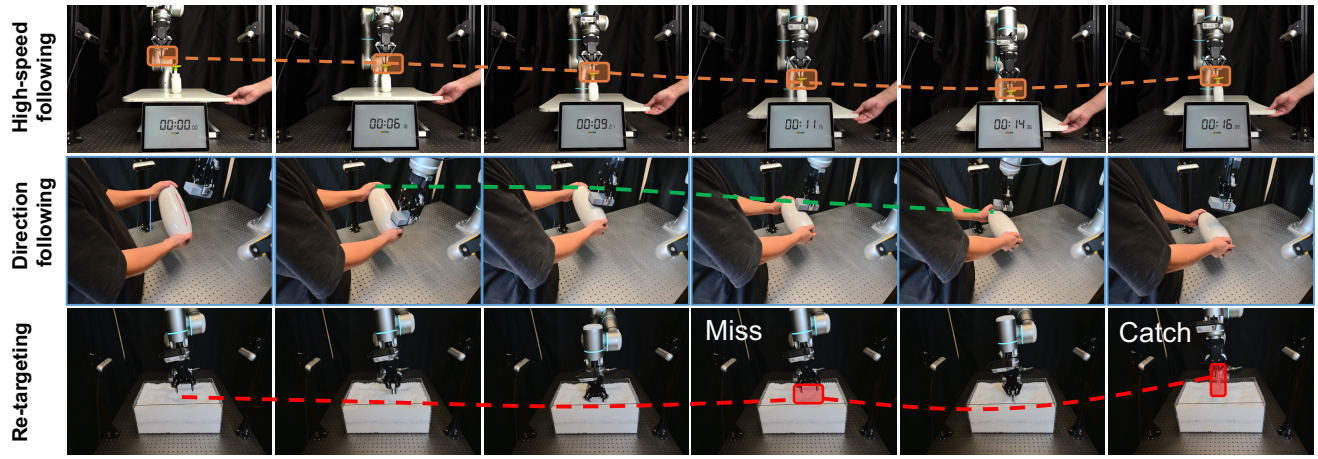


Figure 6. **Additional tests on following and re-targeting.** ForceVLA2 exhibits robust position and orientation following, and in object search tasks, it can still perform successful re-grasps even when visual observations fail.

on force-sensitive tasks such as object search, demonstrating superior force perception. π_0 w/ F reaches 17%, which is similar to the base version, and prior analyzes (TA-VLA, ForceVLA) also suggest that simple concatenation can act as nuisance input and degrade pre-trained VLA representations, instead of providing consistent control-relevant signals. The ACP achieves a success rate of only 16.0%, primarily due to its limited generalization capabilities.

Force-intensive tasks and overload avoidance. As illustrated in Fig. 5, for force-intensive tasks such as clean board and bottle pressing, excessive contact force can easily push the robot into overload. Among the compared methods, ForceVLA2 actively adjusts its interaction forces and successfully completes these tasks, whereas other VLAs lack sufficient reactive adjustment capabilities.

Dynamic force-tracking under sudden perturbations.

Table 1. **Success rates (%) comparison of different methods.** π_0 w/ F: π_0 with native force input. Each value represents the success rate over 20 trials. `Avg.`: the mean performance across all five contact-rich tasks. cyan : main metric. **bold**: best results.

Type	Method	Press bottle	Clean vase	Clean board	Retri. plate	Assem. gears	Avg.
w/o Force	π_0	35.0	20.0	35.0	0.0	0.0	18.0
	$\pi_{0.5}$	45.0	30.0	45.0	15.0	20.0	31.0
w/ Force	ACP	25.0	30.0	25.0	0.0	0.0	16.0
	π_0 w/ F	30.0	25.0	20.0	10.0	0.0	17.0
	ForceVLA	70.0	25.0	55.0	15.0	10.0	35.0
	ForceVLA2	**80.0**	**75.0**	**70.0**	**35.0**	**70.0**$^{\uparrow 50}$	**66.0**

Table 2. **Ablation Study of Different Modules in the ForceVLA2 Model.** `FP`: Force Prompt. `ME`: Multimodal Encoder. `CM`: Cross-Scale MoE module. Entries indicate success rate (%). gray : baseline results.

FP	ME	CM	Press bottle	Clean vase	Clean board	Retri. plate	Assem. gears	Avg.
			35.0	20.0	35.0	0.0	0.0	18.0
✓			60.0	25.0	40.0	5.0	5.0	27.0$^{\uparrow 9}$
✓	✓		60.0	40.0	65.0	5.0	30.0	40.0$^{\uparrow 13}$
✓	✓	✓	**80.0**	**75.0**	**70.0**	**35.0**	**70.0**	**66.0**$^{\uparrow 26}$

Table 3. **Ablation of input modality combinations in the Cross-Scale MoE module.** `VM`: Visual Modality. `FM`: Force Modality. Entries indicate success rate (%).

VM	FM	Press bottle	Clean vase	Clean board	Retri. plate	Assem. gears	Avg.
		70.0	30.0	60.0	0.0	20.0	36.0
✓		**85.0**	40.0	**85.0**	0.0	40.0	50.0$^{\uparrow 14}$
✓	✓	80.0	**75.0**	70.0	**35.0**	**70.0**	**66.0**$^{\uparrow 16}$

In addition to outperforming baselines on standard benchmarks, ForceVLA2 exhibits exceptional force-tracking performance across a variety of dynamic tests. As depicted in Fig. 6, during the bottle pressing task, when the robot arm is about to press the bottle, we abruptly lower the base. Thanks to its force-aware interaction, ForceVLA2 reacts quickly to the new contact configuration and completes the pressing. In contrast, other VLAs slowly chase the new EE 6D pose, leading to failure to maintain stable contact.

Robust wiping and re-targeting in object search. In the vase wiping task, when the vase is rotated such that its surface normal becomes misaligned with the arm's original motion direction, ForceVLA2 drives the arm to slide along the vase and move upward, maintaining contact and successfully completing the wiping motion. Other VLAs continue along their original trajectory, which often results in arm overload and task failure. In the Retrieve Plate task, ForceVLA2 explores the sandbox through force-guided probing; even after an initial failed grasp, it autonomously retries, showcasing robust adaptation.

5.3. Ablation Study

Ablation setup. In this section, we progressively introduce the additional modules built on top of π_0, including the force prompts, the Cross-Scale MoE module, and the Multimodal Encoder, and compare their impact across multiple tasks. In particular, we conduct an ablation on the Cross-Scale MoE module by varying its modality inputs and outputs to verify the effectiveness of our design. Additional experiments focusing on the Multimodal Encoder are provided in the Appendix C.

Component-wise ablations on FP, CM, and ME (Q2). As shown in Tab. 2, we conduct a stepwise ablation in which we progressively add the Force Prompt (FP), Cross-Scale MoE (CM), and Multimodal Encoder (ME) modules on top of π_0, and measure the success rate over five tasks. The results show a consistent increase in success rate as more modules are introduced, confirming the effectiveness of our overall architectural design. We can see that the Cross-Scale MoE module provides the most significant gain in success rate among all added components.

Modality fusion in the Cross-Scale MoE module (Q3). We further perform an ablation study on the Cross-Scale MoE module to validate our design choices. By selectively enabling or disabling force modality (FM) and visual modality (VM), we examine whether the proposed cross-scale fusion scheme is indeed necessary and beneficial. As reported in Tab. 3, VM and FM inputs contribute roughly equally within the Cross-Scale MoE module, indicating that both visual–language context and force signals are essential for achieving strong performance. Interestingly, for the Press Bottle and Clean Board tasks, introducing an extra force token on the input side slightly degrades performance. We hypothesize that this effect arises because the force requirements in these tasks are relatively simple, requiring only a single switch between position and force control, so additional force tokens introduce unnecessary degrees of freedom that can perturb an otherwise stable policy, leading to a minor drop in success rate.

6. Conclusion

We present ForceVLA2, a force-aware vision-language-action framework with hybrid force–position control for contact-rich manipulation, and evaluate it on our ForceVLA2-Dataset of five real-world tasks. Across these tasks, ForceVLA2 achieves a success rate of 66%, outperforming ACP, π_0, $\pi_{0.5}$, and ForceVLA, and surpassing the second-best method by 50 percentage points on the most force-sensitive assembling gears task. The model also consistently reduces failures caused by arm overload and unstable contact, and ablation studies show monotonically improved performance as force prompts, the Cross-Scale MoE, and multimodal fusion are added, with the full ForceVLA2 variant achieving the best results. Overall, ForceVLA2 coupling force–position control achieves more reliable and adaptable contact-rich manipulation than existing state-of-the-art VLA baselines.

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